Paper:

Data Augmentation Using Generative Adversarial Networks for Multi-Class Segmentation of Lung Confocal IF Images

Daiki Katsuma*, Hiroharu Kawanaka*, V. B. Surya Prasath,***, and Bruce J. Aronow**,***

*Graduate School of Engineering, Mie University
1577 Kurima-machiya, Tsu, Mie 514-8507, Japan
E-mail: 420M215@m.mie-u.ac.jp
**Division of Biomedical Informatics, Cincinnati Childrens Hospital Medical Center
3333 Burnet Aveue, Cincinnati, OH 45229, USA
***Department of Pediatrics, University of Cincinnati College of Medicine
Cincinnati, OH 45257, USA
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The human lung is a complex organ with high cellular heterogeneity, and its development and maintenance require interactive gene networks and dynamic cross-talk among multiple cell types. We focus on the confocal immunofluorescent (IF) images of lung tissues from the LungMAP database to reveal lung development. Using the current state-of-the-art deep learning-based model, the authors consider obtaining accurate multi-class segmentation of lung confocal IF images. One of the primary bottlenecks in using deep Convolutional Neural Network (CNN) models is the lack of availability of large-scale training or ground-truth segmentation labels. Then, we implement the multi-class segmentation with Generative Adversarial Network (GAN) models to expand the training dataset, improve overall segmentation accuracy, and discuss the effectiveness of created synthetic images in the segmentation of IF images. Consequently, experimental results indicated that 15.1% increased the accuracy of six-class segmentation using Mask R-CNN. In particular, the accuracy of our few data was mainly improved by using our proposed method. Therefore, the synthetic dataset can moderate the imbalanced data and be used for expanding the dataset.

Keywords: immunofluorescence image, segmentation, data augmentation, image synthesis, generative adversarial networks

1. Introduction

Lung cancer is one of the leading causes of cancer deaths in the world, and in particular, in the United States, the annual mortality number is in excess of 150,000 [1]. Studying the lung structures that perform gas exchange function is important in identifying disease mechanisms. This task looks simple; however, the human lung is a complex organ with high cellular heterogeneity. Its development and maintenance require interactive gene networks and dynamic cross-talk among multiple cell types. Furthermore, it is difficult to understand various causes of abnormal lung developmental patterns and adult lung disorders at the molecular level until we understand its normal development [2]. Quantification of normal lung development/formation and gene/cellular/molecular levels requires carefully assessing various lung imaging modalities.

The LungMAP consortium is one of the research projects for lung development. The consortium was initiated by the National Institutes of Health (NIH) in order to advance molecular, physiologic, and imaging research on the alveolar lung stage using normal, non-diseased human and mouse lung samples [3, 4]. Along with the availability of various types of data, such as biological sequencing data, RNA, protein, lipid, signaling, we can also obtain multi-modal imaging data from the LungMAP database. In particular, lung tissue imaging data shows the protein/cell localization in different tissue structures, and these can provide a foundation upon which to build an atlas of the developing lung [5]. In this work, we focus on the confocal immunofluorescent (IF) images of lung tissues stained with specific sets of proteins. IF images are usually used for a proteomics process to understand the protein localization on the sub-cellular level by image annotation [6–9]. With the availability of large-scale lung confocal IF image, automatic machine learning driving image processing tasks such as segmentation of lung structures and quantification are important tasks. The realization of efficient automatic annotations and segmentation using IF images drastically reduce the manual burden among pulmonary specialists. It promises to increase the usability of such imaging data on the analysis of lung development phases.

In this paper, the authors consider obtaining accurate multi-class segmentation of lung confocal IF images using the current state-of-the-art deep learning-based model. One of the primary bottlenecks in using deep Convolutional Neural Network (CNN) models is the lack of availability of training or ground-truth segmentation labels. We implement the multi-class segmentation with gener-

ative adversarial network (GAN) models to expand the training dataset and improve overall segmentation accuracy. To classify various lung tissue classes along with the region of interest, we propose to utilize recent GAN [10–12]. We apply the method to the multi-class segmentation problem of IF images and discuss the effectiveness of created synthetic images in the segmentation.

We organized the rest of the paper as follows. Section 2 introduces recent GAN models in the context of biomedical images generation. Section 3 mentions the experiment material in this paper. Section 4 describes the details of our scheme, i.e., preparation of dataset and synthetic image creation with GAN. Section 5 provides experimental results and discusses the effectiveness of the proposed scheme from the viewpoint of multi-class segmentation. Section 6 concludes our work and describes the paper along with indicating future works.

2. Related Works

GAN is well-known for generating synthetic data close to the distribution of the training set. GAN can mimic the data distribution and synthesize images with unprecedented realism, opening up a new way to bridge the gap between supervised learning and image generation. It is worthwhile to explore effective imbalanced learning methods because imbalanced data is prevalent in many applications area in the industry, where anomaly detection is critical like electricity pilferage, fraudulent transactions in banks, identification of rare diseases, etc. [13].

In the field of medical imaging, many methods based GAN were applied to a variety of different modalities such as MRI, CT, OCT, chest X-Ray, Dermoscopy, Ultrasound, PET, and Microscopy to implement applications: synthesis, reconstruction, detection, denoising, registration, classification and segmentation [14]. In another research of image reconstruction in the histopathological cancer diagnosis, GAN was applied to transform Hematoxylin and Eosin (H&E) stained images into Immunohistochemistry (IHC) stained images facilitating virtual IHC staining on the same slide. This attempt will contribute to overcoming the limitation of the image analysis and cut off the staining cost [15]. For instance, Gong et al. discussed style consistent image generation for nuclei instance segmentation. In this paper, they tried to generate H&E stained pathological images and discussed the segmentation accuracy of Mask R-CNN [16]. Ranjan et al. proposed a new dual deep generative method for synthesizing human cell protein images using the GAN technique. Also, they evaluated the quality of generated images by the proposed method [17]. Hou et al. also tried to make synthetic images with GAN, and U-net-based approaches were used for segmentation [18].

These technologies, however, have not been investigated deeply for lung analysis. The main point of this paper is how the synthesized images contribute to improving the performance of multi-class segmentation for development analysis with IF images.

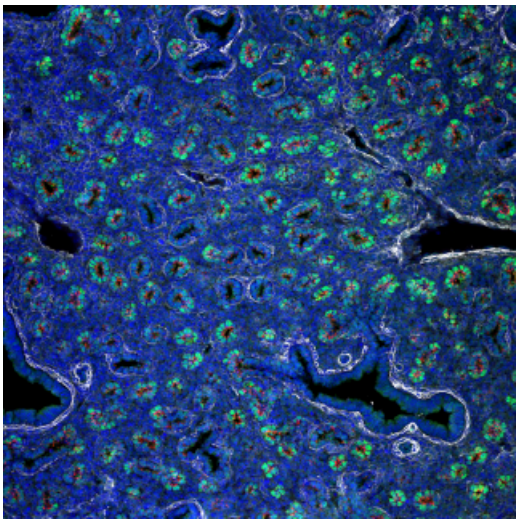


Fig. 1. Example of lung confocal IF image. Three types of proteins are stained as red, green, and white. In this example, red, green, and white pixels correspond Sftpc, Sox9, and Acta2, respectively.

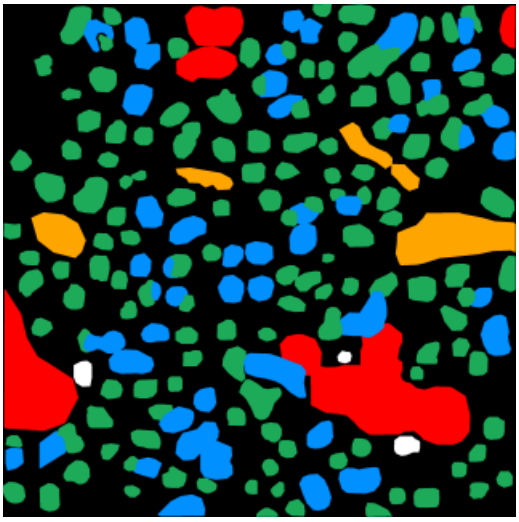


Fig. 2. Example of labelled image. We consider 6 different classes of lung tissues, i.e., background (black: C1), conductive airway (red: C2), distal acinar tubule bud (green: C3), proximal acinar tubule (blue: C4), artery (white: C5), and vein (orange: C6), respectively.

3. Materials

We collected the IF images of the mouse (*Mus musculus* from embryo 16.5 days to postnatal 28 days) and human (*Homo sapiens* from 9 months old to 4 years old) lung from the official website of LungMAP project (https://lungmap.net/). Three types of proteins are stained as red, green, and white in each IF image like **Fig. 1**. The combination of stained proteins depends on each image. Furthermore, we also used the label image like **Fig. 2**, which could be classified into six tissue classes, i.e., Background (black: C1), Conductive Airway (red:

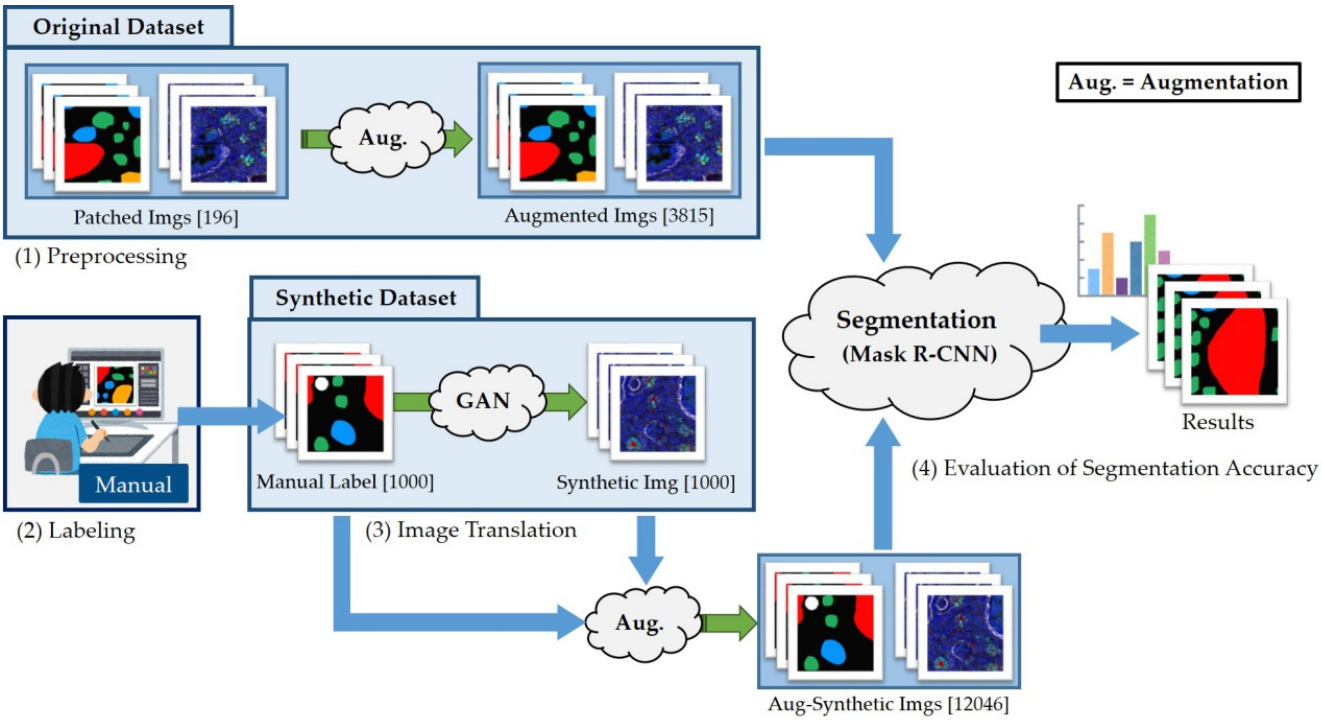


Fig. 3. Workflow in our proposed method. (1) We conducted image preprocessing with traditional augmentation. (2) We drawing a label with a painting tool. (3) GAN translated the label into the synthetic IF image. (4) Segmentation model was evaluated by comparing experimental conditions. The number of images is in brackets.

C2), Distal Acinar Tubule Bud (green: C3), Proximal Acinar Tubule (blue: C4), Artery (white: C5), and Vein (orange: C6).

4. Proposed Method

This paper creates synthetic images with the following method, and the created images are used for expanding the learning dataset for multi-class segmentation. **Fig. 3** illustrates the scheme of our method. The method mainly consists of four processes, i.e., Preprocessing, Labeling, Image Synthesis, and Segmentation. The following subsections describe the detail of each process.

4.1. Preprocessing

First of all, preprocessing is applied to the IF images to prepare the baseline dataset, which is only used for the traditional augmentation. In this paper, the original images were resized 2475×2475 pixels to 1024×1024 pixels because the size of given tissues should be matched to the patch image size for deep learning. Subsequently, we cropped 256×256 pixels with 128 pixels stride, and applied traditional augmentation such as rotating (0°, 90°, 180°, 270°), flipping, random erasing, blurring, sharpening. Consequently, we prepared 3815 images, and this dataset was used as the baseline for learning.

Table 1. Number of objects and average size (256×256 px).

Object [# of images]	Label dataset: Original image [196] # of obj.	Original image [196] Size	Synthetic image [1000] # of obj.	Synthetic image [1000] Size
C1	–	–	–	–
C2	177	9530	628	11515
C3	1978	1123	6449	1384
C4	767	2385	4108	1779
C5	60	843	657	1071
C6	88	4739	295	5891

4.2. Labeling

Before utilizing GAN-based image synthesis, we created the label image manually. In this process, label images that have six types of labels are manually created with a painting tool. **Table 1** shows the numbers of objects and their average sizes before applying traditional augmentation. In the table, object C1 means background so that the number of objects and their sizes are described as "–."

Also, we prepared the label images considering the positional relations of each tissue. For instance, a conductive airway, which is expressed by a red label (C2), is usually located near an artery (white label: C5). Such knowledge is very important in the labeling process because if the created label is not based on topological knowledge, a created image will not reflect features of the original im-

Table 2. Number of objects and pixels in train dataset (256 × 256 px).

Object [# of images]	Original dataset [3815]								Synthetic dataset [12046]	
	Dataset-1 [914]		Dataset-2 [992]		Dataset-3 [929]		Dataset-4 [980]			
	# of obj.	Pixels	# of obj.	Pixels	# of obj.	Pixels	# of obj.	Pixels	# of obj.	Pixels
C1	–	28.8M	–	34.5M	–	36.4M	–	33.0M	–	492.1M
C2	1140	13.3M	807	4.7M	603	4.6M	886	9.3M	7683	83.1M
C3	7029	7.6M	11386	12.3M	10617	11.3M	9402	10.8M	76901	102.6M
C4	3538	7.5M	4055	12.4M	3649	6.5M	3621	8.1M	49074	83.5M
C5	385	256k	260	233k	221	263k	281	182k	7838	8.1M
C6	352	2.1M	332	840k	459	178k	648	2.8M	3606	19.9M

Table 3. Number of objects and pixels in test dataset (1024 × 1024 px).

Object	Dataset-1		Dataset-2		Dataset-3		Dataset-4	
	# of obj.	Pixels	# of obj.	Pixels	# of obj.	Pixels	# of obj.	Pixels
C1	–	473k	–	530k	–	614k	–	535k
C2	7	261k	10	103k	4	84k	5	144k
C3	91	131k	130	200k	136	199k	113	186k
C4	43	150k	37	200k	43	114k	41	138k
C5	7	6.6k	4	5.0k	3	3.8k	4	30k
C6	2	28k	2	9.5k	4	34k	4	43k

age. To prevent this, we prepared the label image based on topological information.

4.3. Image Translation Using GAN

We utilized Pix2Pix, Pix2PixHD, GauGAN for image translation [10–12]. These models can be trained with all images prepared in Section 4.1 and generate a synthetic image from a given label. We focused on this operation and considered the synthetic image and the label as a pair image set for the segmentation dataset. The pair of images is applied to the segmentation dataset as a ground truth and training image.

Pix2Pix has a generator that uses skip connections, and its discriminator is designed to look at the patches rather than the entire image to distinguish real from synthetic images [10]. This, in turn, helps the generator to create high-level details associated with real data distribution. Pix2PixHD is a method based on the Pix2Pix. This method stabilizes learning at high resolution and enables the generation of details. Moreover, it embedded the instance-level object semantic information to improve the image quality [11]. GauGAN is a state-of-the-art image synthesis model that focuses on converting segmentation masks to photorealistic images. It can better preserve semantic information against common normalization layers to propose a new convolutional normalization method called spatially-adaptive denormalization (SPADE) [12]. Thus, this normalization layer tends to keep the information contained in the semantic input masks.

It is easily expected that the quality of synthetic images generated by newer generation models is higher than other models because these models have effective architectures for image translation. However, we also aim to get any clue for an effective data augmentation approach by comparing different models and conditions.

4.4. Evaluation of Segmentation Accuracy

In this scheme, Mask R-CNN, which was one of the well-known segmentation models in the field of medical image segmentation, was used as a segmentation model [19]. We implemented Mask R-CNN and tuned the parameters of the model by changing the number of synthetic images.

Before the evaluation experiments, we prepared 1000 synthetic images for segmentation detailed in **Table 1**. We also applied traditional augmentation to the synthetic images. Consequently, 3815 original images and 12046 synthetic images were used for the experiments. The augmented data conditions are shown in **Table 2**.

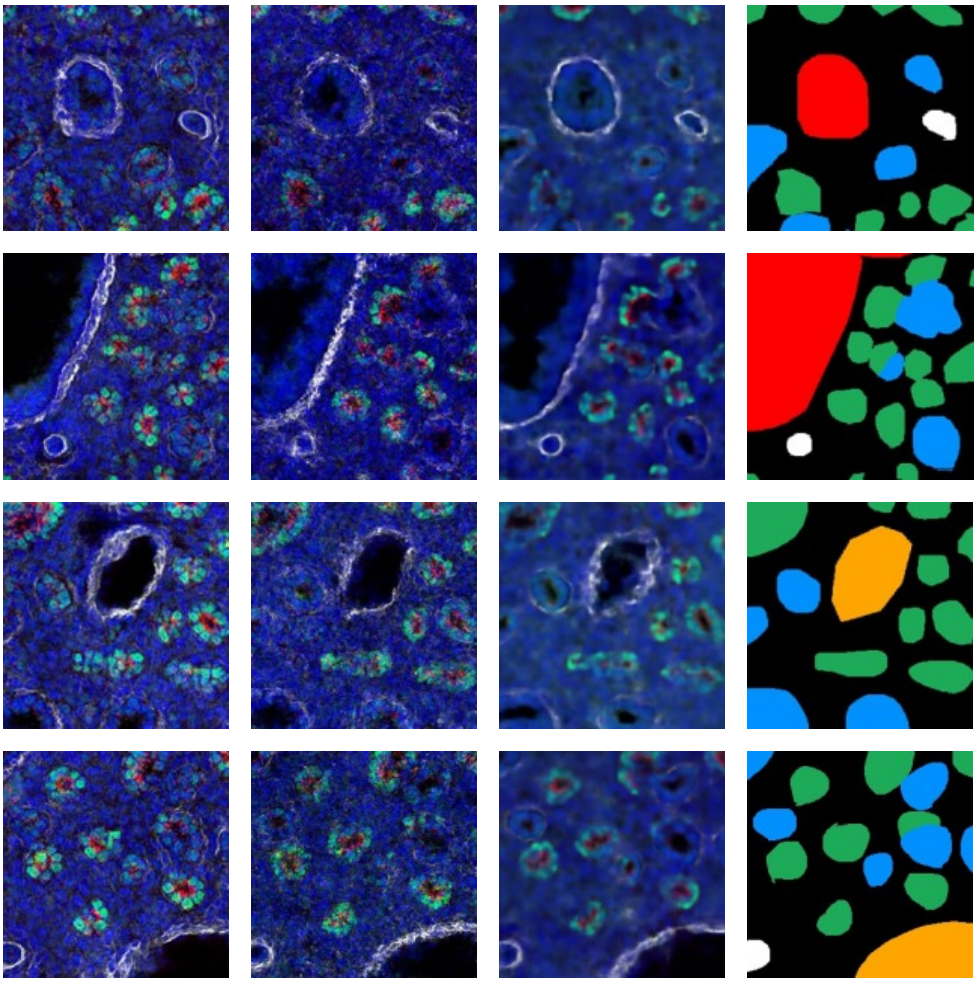
In the learning process, we conducted the 4-fold cross-validation to divide only the original datasets. Divided one was used as a validation dataset, and the others were used as train datasets detailed in **Table 2**. The test dataset was prepared before image patching and augmentation, and its condition is shown in **Table 3**.

In the test process, the segmentation model predicts an image whose size is 256 × 256 pixels, and then the obtained image is stitched into the entire image (1024 × 1024 pixels). After stitching the image, we evaluated segmentation performance with the dice coefficient score for class c (D_c), which was defined by

$$D_c = \frac{2\sum_x\sum_y |\mathbf{C}_{output_{xy}} \cdot \mathbf{C}_{gt_{xy}} \cdot \mathbf{U}_c|}{\sum_x\sum_y |\mathbf{C}_{output_{xy}} \cdot \mathbf{U}_c| + \sum_x\sum_y |\mathbf{C}_{gt_{xy}} \cdot \mathbf{U}_c|}, \quad . \tag{1}$$

Table 4. Each model's hyper parameter for learning.

Model	Batch size	Lrate	Momentum		Epoch	Optimizer
Pix2Pix	8	0.002	beta1: 0.9	beta2: 0.999	990	adam
Pix2PixHD	8	0.0002	beta1: 0.5	beta2: –	200	adam
GauGAN	1	0.0002	beta1: 0.0	beta2: 0.9	50	adam
MRCNN	2	0.001	beta1: 0.9	beta2: –	50	SGD



(a) GauGAN (b) Pix2PixHD (c) Pix2Pix (d) Label Image

Fig. 4. Examples of synthetic images we create. After generated, these images are utilized as a pair of images for training segmentation model. In the figure, (a), (b), (c), (d) mean obtained results by GauGAN, Pix2PixHD, Pix2Pix, and Label, respectively.

where $\mathbf{C}_{output_{xy}}$ is predicted class vector at any pixel (x,y) in the image, and $\mathbf{C}_{gt_{xy}}$ is ground truth class vector as a one-hot vector. In the above equation, $\mathbf{U}_c$ means the one-hot vector for class c, and it is defined by the following equation.

$$\mathbf{U}_c = (U_1, \ldots, U_N) \text{ where } U_i = \begin{cases} 0 & (i \neq c) \\ 1 & (i = c) \end{cases}, \quad c = 1, 2, \ldots, N. \quad (2)$$

In Eq. (2), c means the class, N is the number of classes. In the testing process, we used only the original images for the evaluation. The distribution of objects in each testing dataset is shown in **Table 3**.

5. Experimental Results and Discussion

5.1. Created Synthetic IF Images by GANs

The GAN models used in this paper were trained with the original dataset and applied for the image translation from the label images. The each hyper parameter for learning was shown in **Table 4**. Almost parameter were not changed in the public domain system described in [10–12, 19]. In the experiment, we created all synthetic images by using each GAN model. **Fig. 4** shows examples of created synthetic images by GAN models and label images. These results show that all GAN models could create synthetic IF images from the given label images, and the created images reflect features of the original images

Table 5. Comparison with segmentation accuracy.

GAN	Training set	C1	C2	C3	C4	C5	C6	Average
	Original	0.837	0.717	0.809	0.625	0.104	0.642	0.622
Pix2Pix	6000	0.862	0.792	0.806	0.737	0.527	0.784	0.752
	9000	0.869	0.772	0.817	0.684	0.594	0.843	0.763
	All	0.866	0.809	0.811	0.710	0.652	0.789	**0.773**
Pix2PixHD	6000	0.872	0.781	0.815	0.716	0.625	0.707	0.753
	9000	0.870	0.783	0.809	0.685	0.629	0.762	0.756
	All	0.872	0.779	0.816	0.689	0.626	0.805	**0.765**
GauGAN	6000	0.871	0.796	0.809	0.720	0.614	0.785	0.766
	9000	0.863	0.807	0.804	0.704	0.550	0.798	0.754
	All	0.867	0.833	0.804	0.708	0.632	0.765	**0.768**

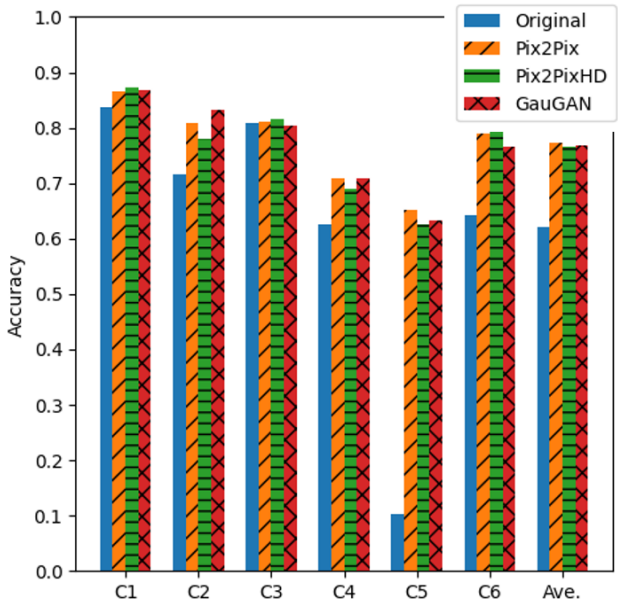
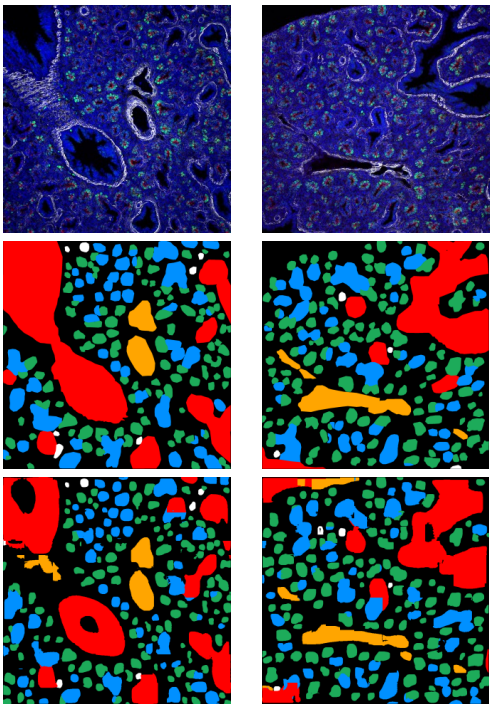


Fig. 5. Accuracy graph to compare the each result of additional 12000 synthetic images and non-addition dataset.



(a) Case 1 (b) Fail case 2

Fig. 6. Examples of failures in segmentation with our approach.

well.

In the case of Pix2Pix, the created images lost focus compared to others, i.e., GauGAN and Pix2PixHD. Pix2Pix is an old model compared to others, and its architecture does not support high resolution. This is why the results were obtained. However, the created IF images look like they have enough capability to be used as a dataset not only for segmentation but also for image classification.

5.2. Evaluation of Segmentation Accuracy

Table 5 and **Fig. 5** show the segmentation accuracy of Mask R-CNN. This table summarizes the dice coefficient score for each tissue when the number of added synthetic images are 0 (i.e., only original dataset), 6000, 9000, and all synthetic images, respectively. The rightmost column means the average of segmentation performance in each condition. The obtained results indicate that the accuracy of six class segmentation using Mask R-CNN was improved in all cases. In particular, the segmentation accuracy was increased by 15.1% when 1000 additional synthetic images were used for learning of Pix2Pix. As mentioned before, the synthetic images created by Pix2Pix lost focus compared to others, but these images also contributed to improving the segmentation accuracy of Mask R-CNN. In the field of medical imaging, it is generally difficult to collect plenty of data for machine learning. In particular, deep learning requires several thousands of images for accurate recognition. By using this approach, we do not need to collect plenty of data and only prepare label images considering features of the baseline dataset. **Fig. 6** and **Table 6** show examples of failures in the segmentation process. In the figure, these images mean the original IF images, ground truth, and segmentation result from the first-tier image. In these cases, Mask R-CNN could not segment red (conductive airway: C2) and orange (vein:

Table 6. Confusion matrices in fail cases.

(a) Fail case 1

	C1	C2	C3	C4	C5	C6	Sum of Pred
C1	433213	66296	26443	31059	2035	4077	563123
C2	3871	183311	0	10901	193	0	198276
C3	16979	0	101199	9200	26	0	127404
C4	18149	2586	3195	98642	0	0	122572
C5	184	245	0	0	4330	0	4759
C6	565	8082	0	0	0	23795	32442
Sum of GT	472961	260520	130837	149802	6584	27872	1048576

(b) Fail case 2

	C1	C2	C3	C4	C5	C6	Sum of Pred
C1	456470	20981	26839	27755	349	12607	545001
C2	20945	116653	0	4557	152	0	142307
C3	20883	0	151074	17362	0	0	189319
C4	24583	6280	7605	88683	621	0	127772
C5	881	0	0	0	1918	0	2799
C6	11100	128	0	0	0	30150	41378
Sum of GT	534862	144042	185518	138357	3040	42757	1048576

C6) objects appropriately. Both objects are a type of muscle fibers and their feature, e.g., shapes and distributions of pixels, is similar to each other. Also, in **Table 6**, we can indicate the same considerations above. The table shows that Mask R-CNN still failed to determine the object's boundary ideally. This is one of the reasons for such experimental results being obtained. It is, however, a heavy task for us to prepare thousands of label images manually, and we have to investigate a method to automatically create label images, which is based on the features of the original images. Also, the given data was imbalanced, e.g., the number of C5 and C6 objects is small compared to others. An architecture for imbalanced data should also be involved in the improved method. In addition, the error of 1% corresponds to 10 pixels in the case of LungMAP IF images. If high-resolution images are used for image analysis, this small difference may cause misrecognition. At the moment, nobody knows the reasons and bottleneck of the proposed method. We have to investigate what contributed to image segmentation.

6. Conclusion

In this paper, the authors considered obtaining accurate multi-class segmentation of lung confocal IF images using the current state-of-the-art deep learning-based model. One of the primary bottlenecks in using deep convolutional neural network (CNN) models was the lack of availability of training or ground-truth segmentation labels. We implemented Mask R-CNN with GAN models to expand the training dataset and improve overall segmentation accuracy. To classify various lung tissue classes along with the region of interest, we proposed to utilize recent GAN based-models, such as Pix2Pix, Pix2PixHD, and GauGAN. We applied the method to the multi-class segmentation problem of lung IF images, and the accuracy of six-class segmentation using Mask R-CNN was improved in all cases. In particular, the segmentation accuracy was increased by 15.1% when 1000 additional synthetic images were used for learning of Pix2Pix.

In the discussion, we described four contents in our future works. First, we need to attempt how are the best conditions we can prepare, e.g., adding more synthetic images and increasing the number of C5 and C6. Second, we should investigate a method to automatically create label images, which is based on the features of the original images, to avoid overall manual burden. Third, the given data was imbalanced, in particular, the number of C5 and C6 objects are small compared to others. Robust architectures for handling imbalanced data should also be involved in improving the classification results. Finally, to discover the detailed reasons and bottlenecks, we have to investigate what contributed to image segmentation. The authors are currently investigating these points to improve the overall performance of multi-class segmentation of lung structures.

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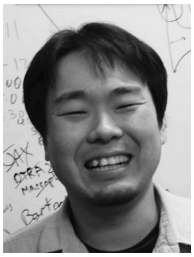
Name:
Daiki Katsuma

Affiliation:
Division of Electrical and Electronic Engineering, Graduate School of Engineering, Mie University

Address:
1577 Kurima-machiya, Tsu, Mie 514-8507, Japan

Brief Biographical History:
2020 Graduated from Department of Electrical and Electronic Engineering, Faculty of Engineering, Mie University
2020- Master Course Student, Graduate School of Engineering, Mie University

Main Works:
• D. Katsuma, H. Kawanaka, B. J. Aronow, and V. B. S. Prasath,"The Effects of Augmentation Using GAN for Confocal Immunofluorescence Image Segmentation," Work-in-Progress Session in 10th Int. Conf. on Informatics, Electronics & Vision (ICIEV2021), 2021.
• D. Katsuma, S. Isaka, H. Kawanaka, B. J. Aronow, and V. B. S. Prasath, "On Data Augmentation Techniques for Deep Learning Multi-class Segmentation of Lung Confocal Immunofluorescent Images," Work-in-Progress Session in 9th Int. Conf. on Informatics, Electronics & Vision (ICIEV), 2020.
• His research area is medical imaging using deep learning techniques, in particular confocal IF images.



Name:
Hiroharu Kawanaka

Affiliation:
Division of Electrical and Electronic Engineering, Graduate School of Engineering, Mie University

Address:
1577 Kurima-machiya, Tsu, Mie 514-8507, Japan

Brief Biographical History:
1999 Graduated from Faculty of Engineering, Mie University
2004 Received Dr.Eng. from Graduate School of Engineering, Mie University
2004 Established Medical Engineering Institute, Inc.
2005 Received NEDO Industry Fellowship and worked for Mie Technical License Organization
2009 Received Ph.D. in Medical Science (DMSc) from Mie University Graduate School of Medicine
2012 Visiting Researcher, Cincinnati Children's Hospital Medical Center
2017 Associate Professor, Graduate School of Engineering, Mie University and Assistant to the President, Mie University

Main Works:
• Y. Matsui, K. Imamura, M. Ooka, S. Chujo, Y. Mase, H. Matsubara, H. Kawanaka, and M. Kondo, "Classification of good visual acuity over time in patients with branch retinal vein occlusion with macular edema using support vector machine," Graefe's Archive for Clinical and Experimental Ophthalmology, doi: 10.1007/s00417-021-05455-y, 2021.
• T. Hayakawa, V. B. S. Prasath, H. Kawanaka, B. J. Aronow, and S. Tsuruoka, "Computational automatic nuclei segmentation methods in digital pathology – A survey," Archives of Computational Methods in Engineering, doi: 10.1007/s11831-019-09366-4, 2019.
• His current research topics are medical informatics, medical document image analysis, graphics recognition, welfare information systems.

Membership in Academic Societies:
• The Institute of Electrical and Electronics Engineers (IEEE)
• The Healthcare Information and Management Systems Society (HIMSS)
• Japan Society for Fuzzy Theory and Intelligent Informatics (SOFT)
• The Institute of Electronics, Information and Communication Engineers (IEICE)

Name:
V. B. Surya Prasath

Affiliation:
Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center
Department of Pediatrics, University of Cincinnati College of Medicine

Address:
3333 Burnet Avenue, Cincinnati, OH 45229, USA
Cincinnati, OH 45257, USA

Brief Biographical History:
2009 Received his Ph.D. in mathematics from the Indian Institute of Technology Madras
2010-2011 Postdoctoral Fellow, Department of Mathematics, University of Coimbra
2012-2015 Postdoctoral Fellow, Computational Imaging and VisAnalysis (CIVA) Lab, University of Missouri
2016-2017 Assistant Research Professor, University of Missouri
2018- Assistant Professor, Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center and Departments of Biomedical Informatics, Electrical Engineering and Computer Science, University of Cincinnati
2011,2012 Summer fellowships/visits at Kitware Inc., The Fields Institute, and IPAM, University of California Los Angeles (UCLA)

Main Works:
• V. B. S. Prasath, R. Pelapur, G. Seetharaman, and K. Palaniappan, "Multiscale structure tensor for improved feature extraction and image regularization," IEEE Trans. on Image Processing, Vol.28, No.12, pp. 6198-6210, doi:10.1109/TIP.2019.2924799, 2019.
• V. B. S. Prasath and D. Vorotnikov, "On time adaptive critical variable exponent vectorial diffusion flows and their applications in image processing I: Analysis," Nonlinear Analysis, Vol.168, pp. 176-197, doi: 10.1016/j.na.2017.11.013, 2018.
• V. B. S. Prasath, "Polyp detection and segmentation from video capsule endoscopy: A review," J. of Imaging, Vol.3, No.1, doi: 10.3390/jimaging3010001, 2017.
• His main research interests include nonlinear PDEs, regularization methods, inverse and ill-posed problems, variational, PDE based image processing, and computer vision with applications in remote sensing, biomedical imaging domains. His current research focuses are in data science, and bioimage informatics with machine learning techniques.

Name:
Bruce J. Aronow

Affiliation:
Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center
Department of Pediatrics, University of Cincinnati College of Medicine

Address:
3333 Burnet Avenue, Cincinnati, OH 45229, USA
Cincinnati, OH 45257, USA

Brief Biographical History:
1986 Received Ph.D. in Biochemistry, University of Kentucky
2006- Full Professor, Cincinnati Children's Hospital Medical Center, University of Cincinnati
2009- Core Director, Center for Modeling Molecular Mechanisms of Disease, part of the Cincinnati Academic Health Centers CTSA
2014- Scientific Director, Center for Computational Medicine, State of Ohio Third Frontier Center of Excellence

Main Works:
• J. Chen, E. E. Bardes, B. J. Aronow, and A. G. Jegga, "ToppGene Suite for gene list enrichment analysis and candidate gene prioritization," Nucleic Acids Research, Vol.37, Issue suppl_2, pp. W305-W311, doi: 10.1093/nar/gkp427, 2009.
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• E. B. Wilson, D. H. Yamada, H. Elsaesser, J. Herskovitz, J. Deng, G. Cheng, B. J. Aronow, C. L. Karp, and D. G. Brooks, "Blockade of chronic type I interferon signaling to control persistent LCMV infection," Science, Vol.340, No.6129, pp. 202-207, doi: 10.1126/science.1235208, 2013.
• His area of research is devoted to understanding mechanisms of health and disease. His group builds databases and carries out analyses of experimental and clinical data to help researchers predict active pathways of cancer based on gene-expression, gene mutation, and biological function analyses. His recent work includes development of a new database that profiles drug sensitivities of different cancers to help predict the most powerful therapeutic agents against different cancers and provides the research community with tools to rank genes, mutations, pathways, regulatory mechanisms, and cells via http://toppgene.cchmc.org, http://toppcluster.cchmc.org, http://toppmir.cchmc.org and http://gataca.cchmc.org, and http://toppcell.cchmc.org for their roles in normal, disease, and oncogenic processes and functions.